

Week 2: Deciding under *fire*

Context Triage, uncertainty, and decisions under scarcity
Engineering Building for *Broken* Environments

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
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**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

When there is not enough time, information, or capacity to treat everyone, what should we do next?

- Who is most at risk, and how do we decide quickly?
- What changes when our information is incomplete or wrong?
- How can a hospital expand, coordinate, and remain accountable under surge?

Deciding Under Fire: Triage, Uncertainty & Care

Lecturer: *Prof. Annie Hartley*

Light

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Outline

- 01 The problem: When need exceeds capacity
- 02 The birth of Triage: *Larrey's "flying ambulances"*
- 03 Uncertainty and reassessment
- 04 Surge, coordination, and the hospital system
- 05 Accountability: neutrality, ethics, and AI

The word “patient” to wait?

Patient

From Latin *patiens*: one who suffers; one who endures.

suffering is a function of time (make equation)

$$\text{Priority} = \frac{\text{Urgency} \times \text{Benefit} \times \text{Confidence}}{\text{Time} \times \text{Resources}}$$

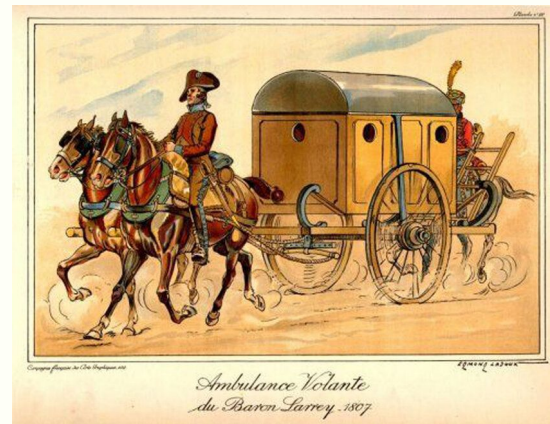
Terminology

Larrey's flying ambulances, 1766-1842

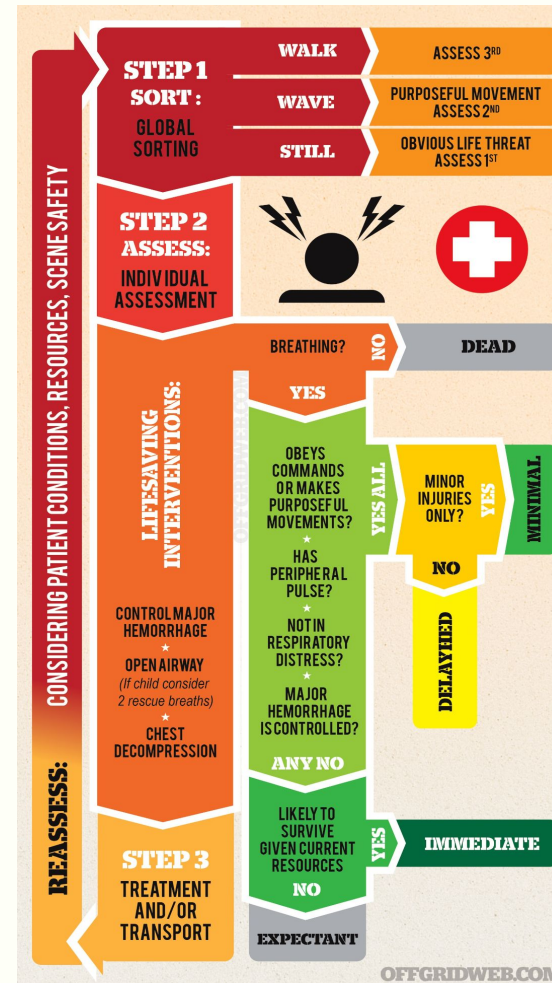
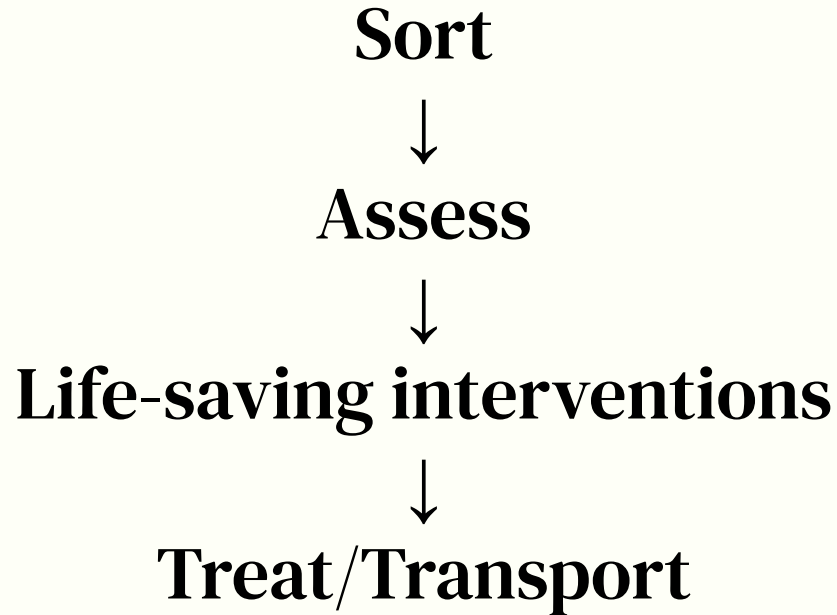
These were mobile medical teams and fast horse-drawn vehicles intended to reach and remove wounded people while fighting was still ongoing.

But what were they optimizing?

- Treat the most severely injured first?
- Treat those most likely to survive?
- Treat those who can return to duty fastest?
- Treat people in arrival order?
- Or preserve resources for patients who appear more salvageable?

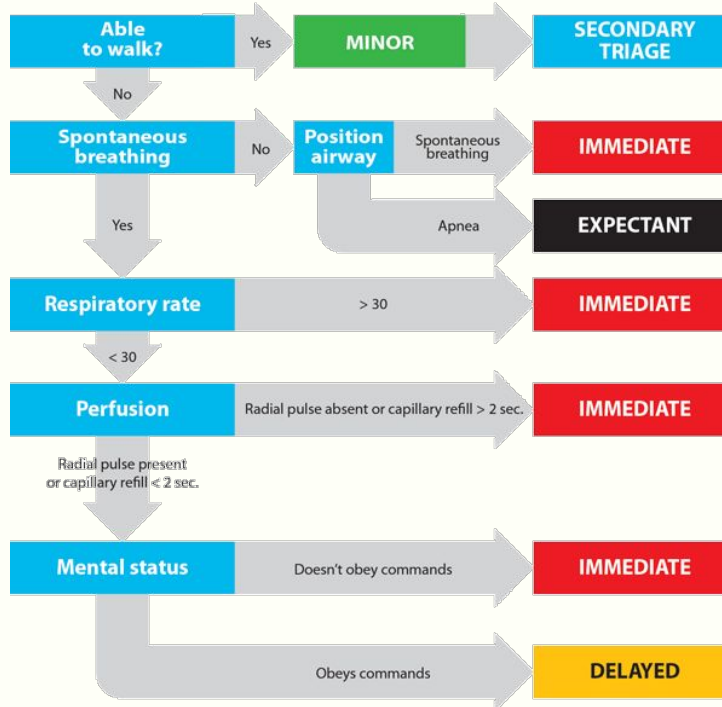


SALT



START - Simple Triage And Rapid Treatment

Figure 1: START method for adults



Terminology

Expectant

Categorization for a patient who is unlikely to survive given the available time and resources

ABCDEG

A — Airway: Can air get in?

B — Breathing: Is oxygen moving into the body?

C — Circulation: Is blood moving around the body? Is there major bleeding?

D — Disability: What is happening neurologically—consciousness, pupils, seizures, blood glucose?

E — Exposure / environment: Look for injuries, rashes, burns, bleeding, temperature problems, while preserving dignity and preventing hypothermia.

Glucose

Referral chain

explain primary → tertiary

Home / community



Community health worker or clinic



Primary care / health centre



District hospital



Regional or specialist hospital



Tertiary referral centre



Back home: rehabilitation, follow-up, long-term care

Level	What it is for	Who you might meet	Typical capabilities
Community	Recognise danger early, prevention, trust, continuity	Community health worker, midwife, volunteer, pharmacist	Health education, vaccinations, antenatal care, basic medicines, identifying who needs referral
Primary care	First-contact care close to daily life	Family physician/GP, nurse, midwife, clinical officer, community nurse	Common illnesses, chronic disease, childhood care, uncomplicated births, basic tests, prevention
Secondary care	Hospital-level diagnosis and treatment	Emergency physician, general surgeon, obstetrician, paediatrician, internist, anaesthetist	X-ray, laboratory, inpatient wards, blood transfusion, emergency surgery, caesarean section
Tertiary care	Complex, specialised, technology-intensive care	Trauma surgeon, neurosurgeon, cardiologist, intensivist, oncologist, neonatologist	ICU, CT/MRI, specialist surgery, dialysis, cancer treatment, advanced neonatal and trauma care
Quaternary care <i>(where it exists)</i>	Rare, ultra-specialised expertise	Transplant, genetic, advanced cancer, complex surgical teams	Transplants, experimental therapies, highly specialised procedures

- SOAP

S —

O —

A —

P —

- SBAR

S — Situation: What is happening right now?

B — Background: What important context do they need?

A — Assessment: What do I think is going on?

R — Recommendation / Request: What do I need from you, and how urgently?

“I’m calling about a 26-year-old with abdominal trauma. His blood pressure is falling despite fluids. He was stable on arrival but is now pale and confused. I am concerned about internal bleeding. I need you in resuscitation now and would like the theatre alerted.”

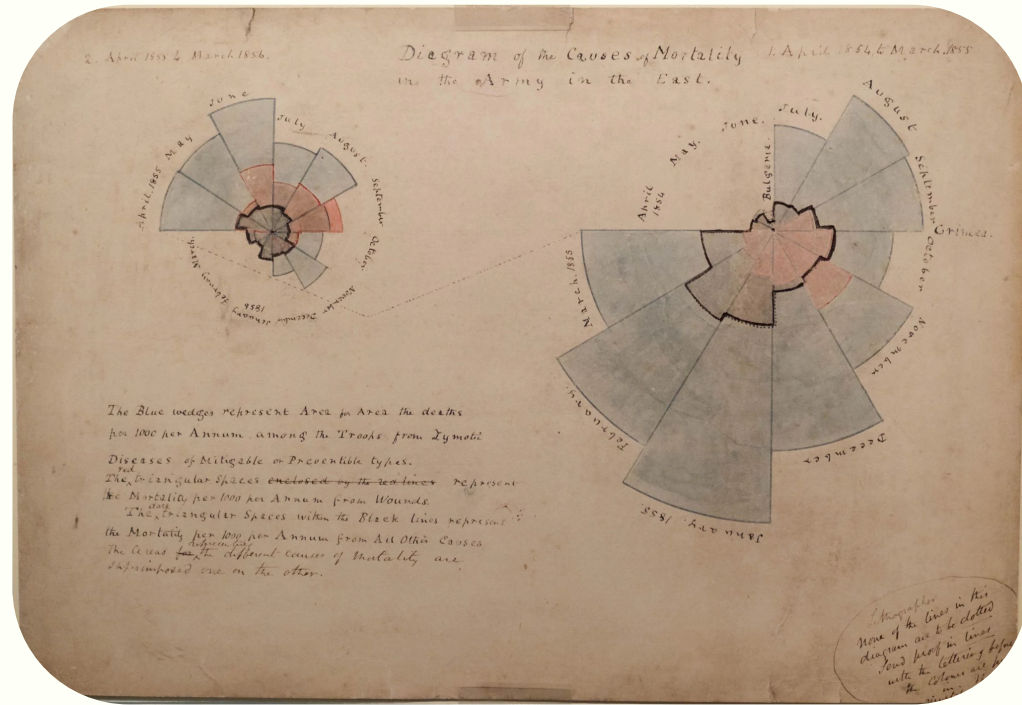
From optimizing patients to optimizing the system

Florence Nightingale got organized

Not just about more beds or quicker transport...

Transport to WHERE

People die of other things when they are here



Pre-positioning

UNICEF pre-made boxes that ship out for diff disasters

[UNICEF Supply Division Warehouse Tour](#)

Menstrual hygiene kits

[UNICEF menstrual hygiene products in emergencies](#)



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In love of checklists

A good checklist does three things:

1. **Protects against stupid-but-preventable omissions**
Did we check identity? Is the equipment present? Has someone said the critical concern aloud?
2. **Creates a pause point**
It forces people to stop at the moments where rushing is dangerous.
3. **Turns individuals into a team**
The best checklists make people introduce themselves, state concerns, and share a plan.

Terminology

WHO
Surgical
Safety
Checklist

SIGN IN
TIME OUT
SIGN OUT

In love of checklists

When complexity exceeds memory, safety depends on making the essential visible, shared, and speakable.

1. **The enemy is not ignorance. It is failure of execution.**
In medicine, people often know the right thing to do. They fail because they are interrupted, rushed, tired, working across silos, or managing too many moving parts.
2. **There are two kinds of failure: “we didn’t know” and “we knew, but didn’t do it.”**
Modern medicine increasingly suffers from the second. The checklist targets that gap.
3. **The checklist is a cognitive prosthetic.**
Like a map, calculator, or microscope, it extends what humans can reliably do. It does not make someone less expert; it frees attention for judgment, surprises, and care.
4. **The important items are often embarrassingly basic.**
Confirm the patient. Check allergies. Give antibiotics. Count the instruments. Ensure blood is available.
The humbling point: lives can turn on things everyone “already knows.”

Terminology

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In love of checklists

When complexity exceeds memory, safety depends on making the essential visible, shared, and speakable.

5. **A good checklist does not try to list everything.**

It is short, specific, and used at natural pause points. If it becomes a bureaucratic encyclopaedia, people stop using it.

6. **The real intervention is communication, not ticking boxes.**

In the surgery checklist, people introduce themselves, name concerns, and say the plan out loud. It creates permission for the nurse, junior doctor, anaesthetist, or technician to interrupt.

7. **Checklists flatten hierarchy at the moment it matters.**

A junior nurse can say, “We have not given antibiotics,” because the checklist—not their status—has made the omission visible.

Terminology

WHO

Surgical
Safety
Checklist

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In love of checklists

When complexity exceeds memory, safety depends on making the essential visible, shared, and speakable.

8. **They turn a collection of experts into a team.**

A room full of brilliant people is not automatically a coordinated system. The checklist gives them a shared script, shared facts, and a moment to synchronize.

9. **They make uncertainty discussable.**

“What is the worst thing that could happen?” is a checklist question. That is incredibly relevant to AI: good systems should expose uncertainty and prompt challenge, not merely issue confident recommendations.

10. **They are a form of respect.**

A checklist says: no patient should be harmed because the team was distracted, exhausted, intimidated, or assumed someone else had done the obvious thing.

Terminology

WHO

Surgical
Safety
Checklist

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TIME OUT
SIGN OUT

In love of checklists

Elderly man has wrong leg amputated at Austrian clinic

Freistadt Clinic apologises for 'tragic mistake' which local media said the patient did not initially recognise because of his illness

U.S. NEWS

Florida doctor faces manslaughter charge for allegedly removing wrong organ during surgery

Terminology

WHO

**Surgical
Safety
Checklist**

SIGN IN
TIME OUT
SIGN OUT

In love of checklists

Gloucestershire Hospitals NHS foundation trust **NHS**

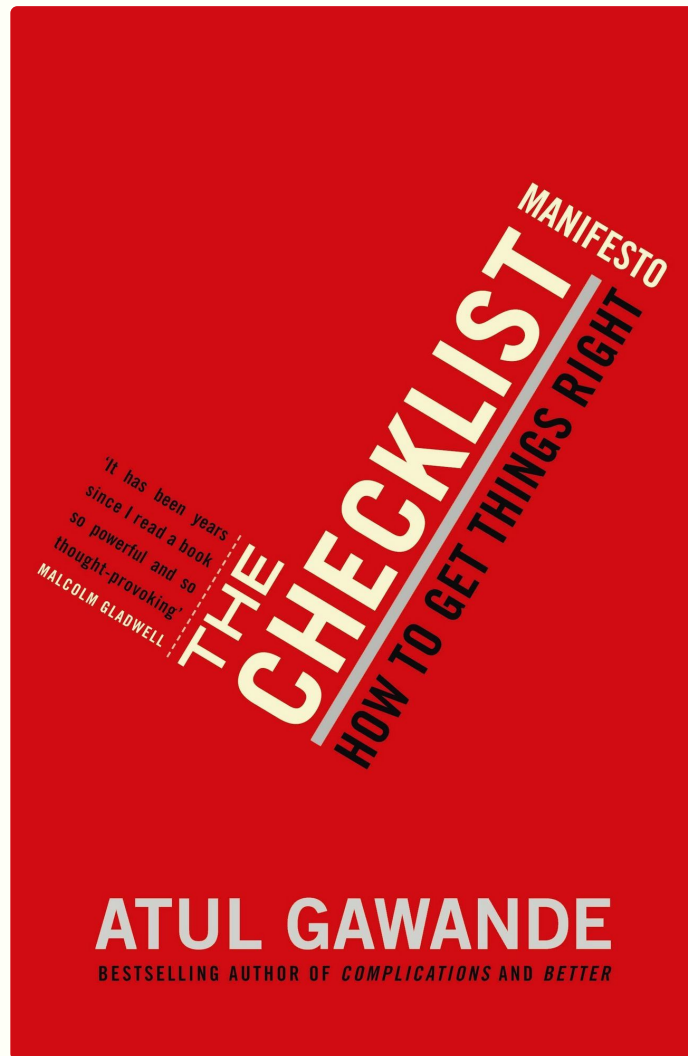
WHO checklist

SIGN IN	TIME OUT	SIGN OUT
Anaesthetist and assistant (Before induction of anaesthesia) Patient confirmation - Name - Procedure - Site marked Allergies? Anaesthetic safety - Airway plan - Antibiotics E-issue blood available? STOP Stop before you block	Introductions Team: Confirm patient's - Name - Procedure - Site and side - Imaging - Allergies Surgeon - Significant blood loss? Team - Diathermy on - Antibiotics given - Specimen plan - Warming on - VTE prophylaxis on Anyone not happy to start?	Whole theatre team - Procedure recorded as... - Counts correct - Specimens labelled - Packs removed / labelled - Lines flushed - Equipment problems Post op plan - Specific concerns? - VTE and antibiotic plan - Daycase?

Terminology

WHO Surgical Safety Checklist

SIGN IN
TIME OUT
SIGN OUT



End

DOMAIN

Questions?

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Building for *Broken* Environments

Lecturer: *Dr. Lars Klein*

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Engineering

This course will be heavily project-driven

Imagine that you're building for people in need, that your engineering affects them immediately

One key term here is **Accountability**

How do we prevent mistakes?

Why does this matter so much?

In the Wild...

Let's assume you're a bio-informatician...

Human genes

Mouse RIKEN IDs

MARCH1
SEPT2
DEC3

2310009E13
0610009E02

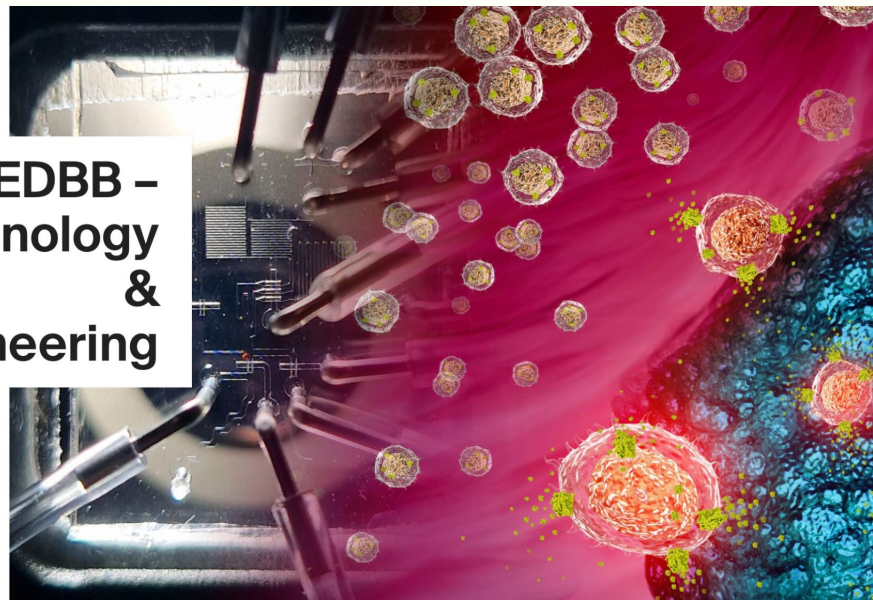
AND
OR
NOT

Drosophila genes

**EDBB –
Biotechnology
&
Bioengineering**

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In the Wild...

Mistaken Identifiers: Gene name errors can be introduced inadvertently when using Excel in bioinformatics [paper](#)

Background *When processing microarray data sets, we recently noticed that some gene names were being changed inadvertently to non-gene names.*

Results *A little detective work traced the problem to default date format conversions and floating-point format conversions in the very useful Excel program package.*

*The **date conversions** affect at least **30 gene names**;*

*the **floating-point conversions** affect at least **2,000** if Riken identifiers are included.*

*These conversions are **irreversible**; the original gene names cannot be recovered.*

In the Wild...

Gene name errors are widespread in the scientific literature [paper](#)

*Of the selected journals, the proportion of published articles with Excel files containing gene lists that are **affected by gene name errors is 19.6 %**. [...]*

Journals that had the highest proportion of papers with affected supplementary files were Nucleic Acids Research, Genome Biology, Nature Genetics, Genome Research, Genes and Development and Nature (>20 %)

In the Wild...

Guidelines for human gene nomenclature [paper](#)

Box 3 Scenarios that may merit a symbol change

...

- **Symbols that affect data handling and retrieval.** For example, all symbols that autoconverted to dates in Microsoft Excel have been changed (for example, *SEPT1* is now *SEPTIN1*; *MARCH1* is now *MARCHF1*); tRNA synthetase symbols that were also common words have been changed (for example, *WARS* is now *WARS1*; *CARS* is now *CARS1*).

[Show less](#) ^

In the Wild...

Gene name errors: Lessons not learned [paper](#)

Although recent changes have been made to human and mouse gene names to prevent conversion to dates [3], it is uncertain whether such changes have propagated through to other species. To assess this, we downloaded all eukaryotic gene names available in Ensembl and imported this into Excel and collected all genes that were converted to dates

The proportion of publications with Excel gene lists that contain errors was 30.9%; substantially higher than previously reported

So now what?

[Commentary blog post from the authors](#)

Our research shows autocorrect errors, particularly in Excel spreadsheets, can also make a mess of gene names in genetic research. We surveyed more than 10,000 papers with Excel gene lists published between 2014 and 2020 and found [more than 30%](#) contained at least one gene name mangled by autocorrect.

This research follows our 2016 study that found [around 20%](#) of papers contained these errors, so the problem may be getting worse.

We believe the lesson for researchers is clear: it's past time to stop using Excel and learn to use more powerful software.

It's past time to stop using Excel and learn to use more powerful software...

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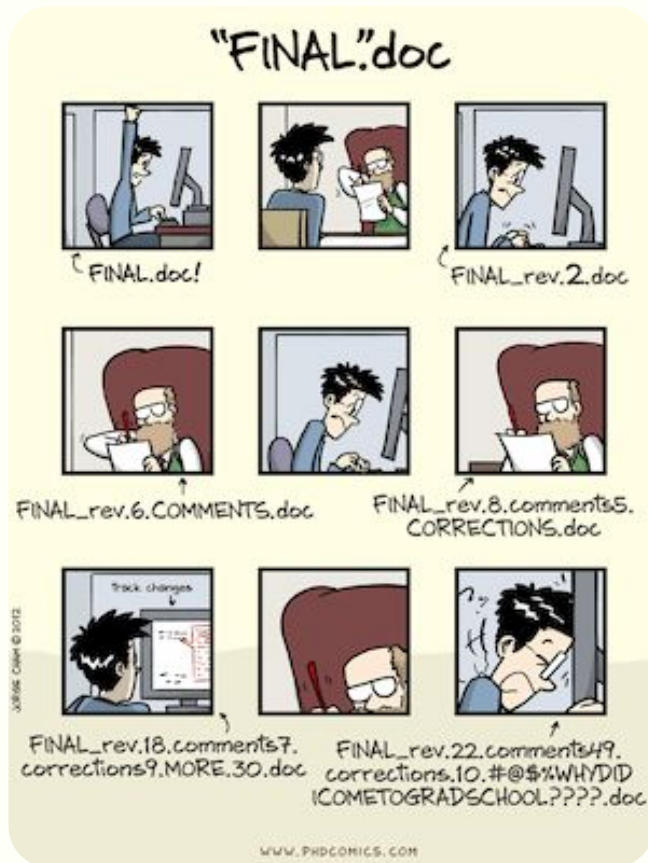
Chances are you already do



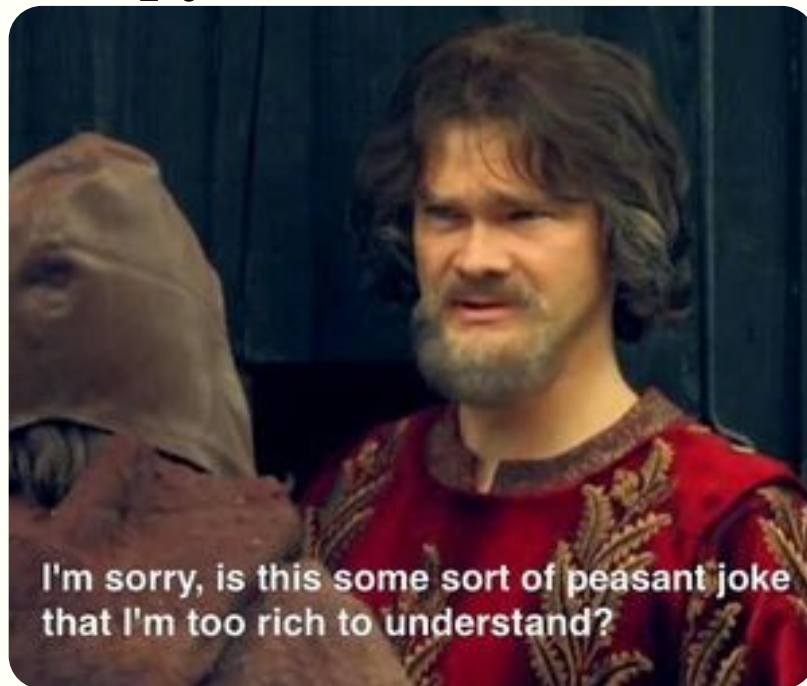
Outline

- 01 Git
- 02 Code Hygiene
- 03 Reproducibility
- 04 Tracking
- 05 Project Time!

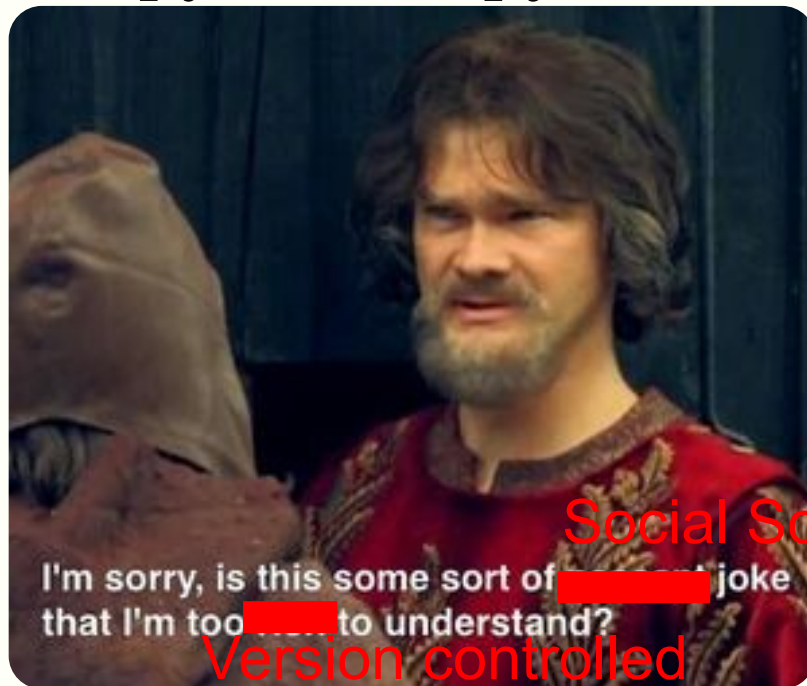
final_v2_ACTUAL.py



final_v2_ACTUAL.py



~~final_v2_ACTUAL.py~~ -> final.py



Social Sciences

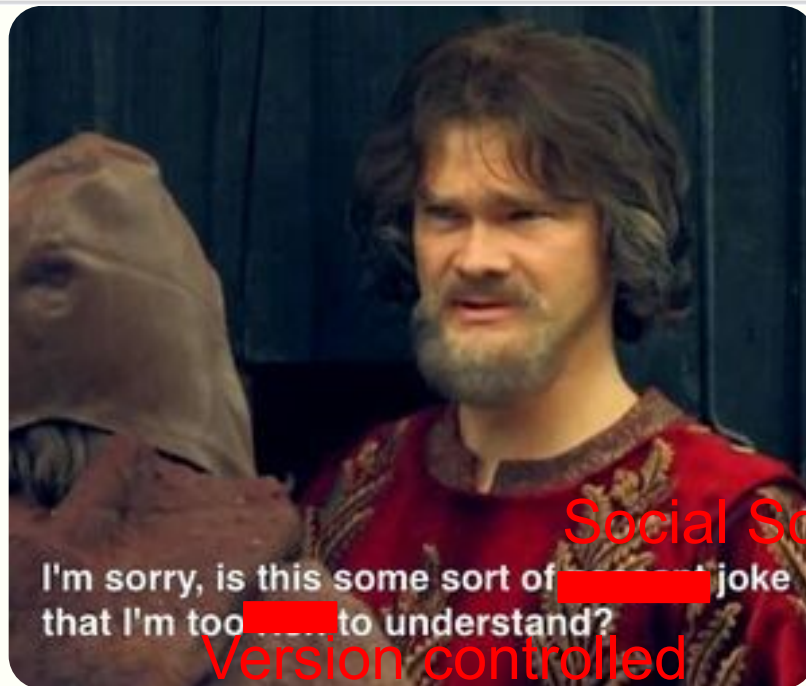
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 final.py

applying feedback from ...

1 minute ago



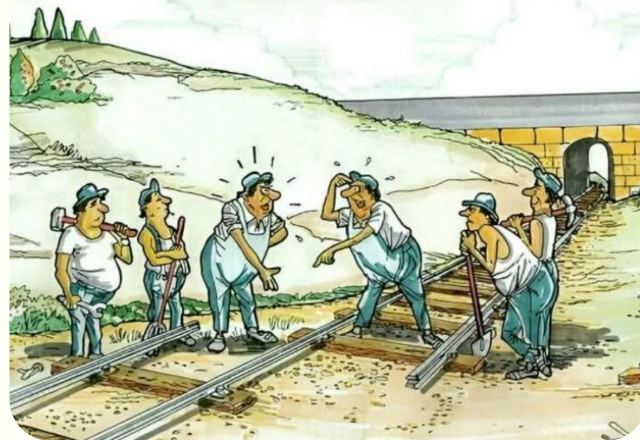
Social Sciences

Git

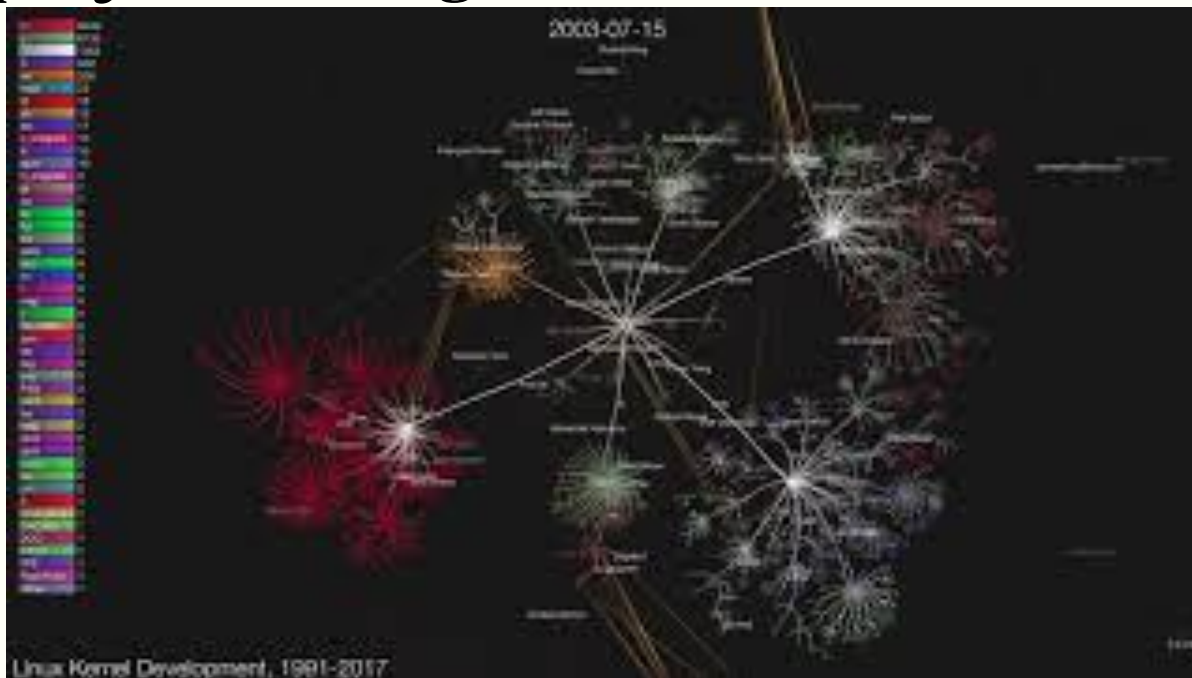
Please ask yourself:

- What is the LARGEST team size you've ever worked in?
- Did that work well?
- How much did you need to coordinate with your teammates?

**"TEAMWORK WITHOUT
COORDINATION LEADS
TO CONFUSION"**



LARGE projects with git ...



Git sanity checklist

All your projects MUST live on git → [BiGHT - Course projects GitHub](#)
This checklist is grade relevant.

Git sanity checklist

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This checklist is grade relevant.

1. Never edit the main branch directly. Development happens on feature branches

Managing a branch protection rule

You can create a branch protection rule to enforce certain workflows for one or more branches, such as requiring an approving review or passing status checks for all pull requests merged into the protected branch.

Who can use this feature?

- 🔑 People with admin permissions or a custom role with the "edit repository rules" permission to a repository can manage branch protection rules.
- 📁 Protected branches are available in public repositories with GitHub Free and GitHub Free for organizations. Protected branches are also available in public and private repositories with GitHub Pro, GitHub Team, GitHub Enterprise Cloud, and GitHub Enterprise Server. See [GitHub's plans](#).

Git sanity checklist

All your projects MUST live on git → [BiGHT - Course projects GitHub](#)
This checklist is grade relevant.

1. Protect the main branch. Development happens on feature branches
2. Branches must be given a semantic name



dev_lars_rangein...



21 Branches

Git sanity checklist

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draft: Dev/lars/fa2layout #16 



Merged

lhk merged 14 commits into `main` from `dev/lars/fa2layout`  on Nov 24, 2025

Git sanity checklist

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This is a shared responsibility. You must empower your teammates to collaborate with you.

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5. Commit messages must be meaningful
6. Data, model weights/checkpoints, .env files, keys don't go in git, write .gitignore on day 1

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```
[(base) lklein@Larss-MacBook-Pro push2whisper % git status  
On branch main  
Your branch is up to date with 'origin/main'.  
  
nothing to commit, working tree clean
```

Code Hygiene

We do expect you to review each other's PRs.

But no one wants to review something like this:

Add black and pre-commit hooks, apply formatting

 `angband-env`

1 parent [3d724a4](#) commit `4e453e9` 

34 files changed +1,630 -859 

Code Hygiene

We do expect you to review each other's PRs.

What was the problem in the previous message?

→ 34 changes files, ~1600 lines of code added, ~800 removed!

You can't possibly expect that people read this.

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Ai coding agents will happily change code back and forth, introduce trivial formatting changes, tweak comments, etc

→ Ultimately touching hundreds or thousands of lines of code for meaningless changes

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Your commits **MUST** be semantic.

Each change is **self contained**, each change is **justified**, each change is **documented**.

Code Hygiene

YAML

 Copy

```
repos:
  - repo: https://github.com/astral-sh/ruff-pre-commit
    rev: v0.16.1
    hooks:
      - id: ruff-check
        args: [--fix]
      - id: ruff-format
```

Reproducibility

Did you ever wonder:

- How did this work?
- What numbers went into this plot?
- Why can I not get the same results again?
- What if patient-facing decisions depend on your numbers?

Reproducibility

Did you ever wonder:

[Nature : 1,500 scientists lift the lid on reproducibility](#)

- How did this work?
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- Why can I not get the same results again?
- What if patient-facing decisions depend on your numbers?

Survey sheds light on the 'crisis' rocking research.

More than 70% of researchers have tried and failed to reproduce another scientist's experiments, and more than half have failed to reproduce their own experiments. Those are some of the telling figures that emerged from *Nature's* survey of 1,576 researchers who took a brief online questionnaire on reproducibility in research.

Reproducibility

So many reasons reproducibility can fail ...

November 19, 2021

Builds Today, Breaks Tomorrow: The Mystery of the Disappearing NPM Dependency

Bella Wiseman, Lead Engineer, Open Source Program Office

Reproducibility

So many reasons reproducibility can fail ...

```
gatsby
|_gatsby-cli
  |_gatsby-recipes
    |_@graphql-tools/schema
      |_value-or-promise
        |_@changesets/cli
          |_tty-table
            |_smartwrap
```



Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

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Easier said than done. For dependencies there's lockfiles.

Reproducibility

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Keep very careful track of all the parameters that go into an experiment!

`package-lock.json` is automatically generated for any operations where npm modifies either the `node_modules` tree, or `package.json`. It describes the exact tree that was generated, such that subsequent installs are able to generate identical trees, regardless of intermediate dependency updates.

This file is **intended to be committed** into source repositories, and serves various purposes:

Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

Easier said than done. For dependencies there's lockfiles.

But how do you track logic, configs, etc?

Reproducibility

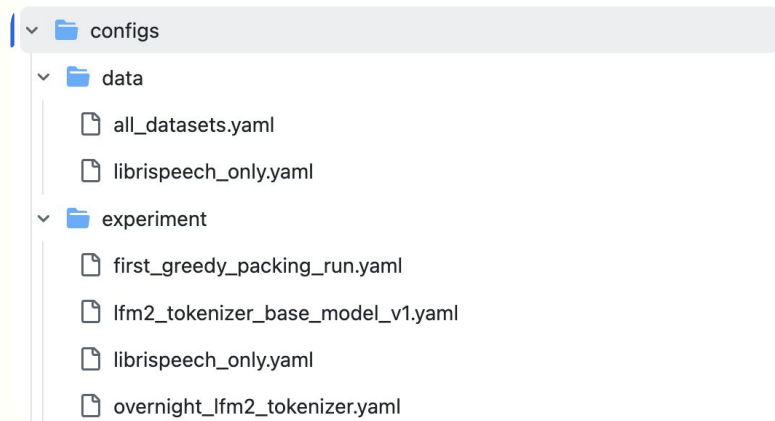
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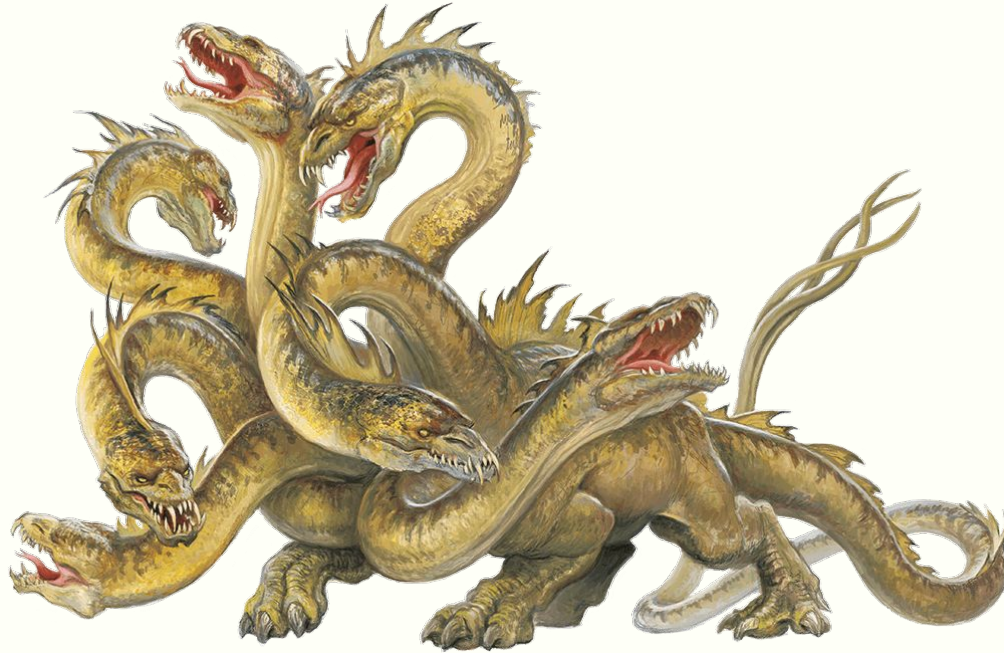
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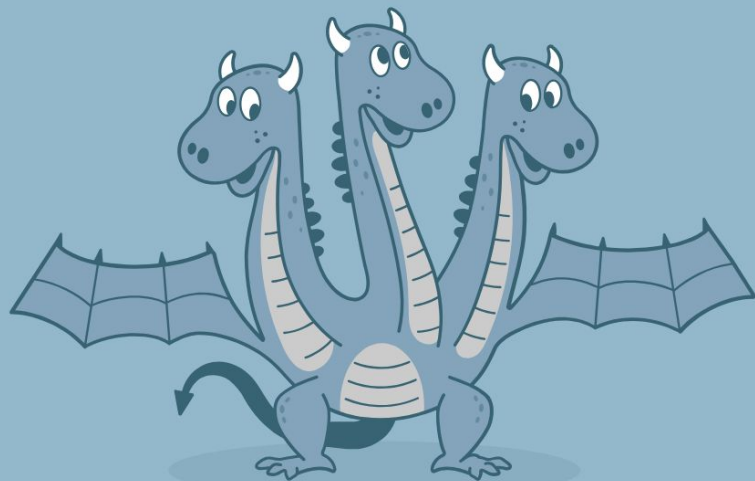
But how do you track logic, configs, etc?



Hydra



Also Hydra (only mildly less scary when first encountered)



Hydra

A framework for elegantly configuring complex applications

Get Started



Star

10,602

Tracking & Logging

If a tree falls in a forest and no one is around to hear it, does it make a sound?

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If code runs on the cluster and no one is around to log it, does it waste compute?

Tracking & Logging

If a tree falls in a forest and no one is around to hear it, does it make a sound?

If code runs on the cluster and no one is around to log it, does it waste compute?

Golden Rule:

You **MUST** account for the resources you use.

→ Every number you produce, **MUST** be reproducible. If it isn't, the compute was wasted!

But how?

W&B

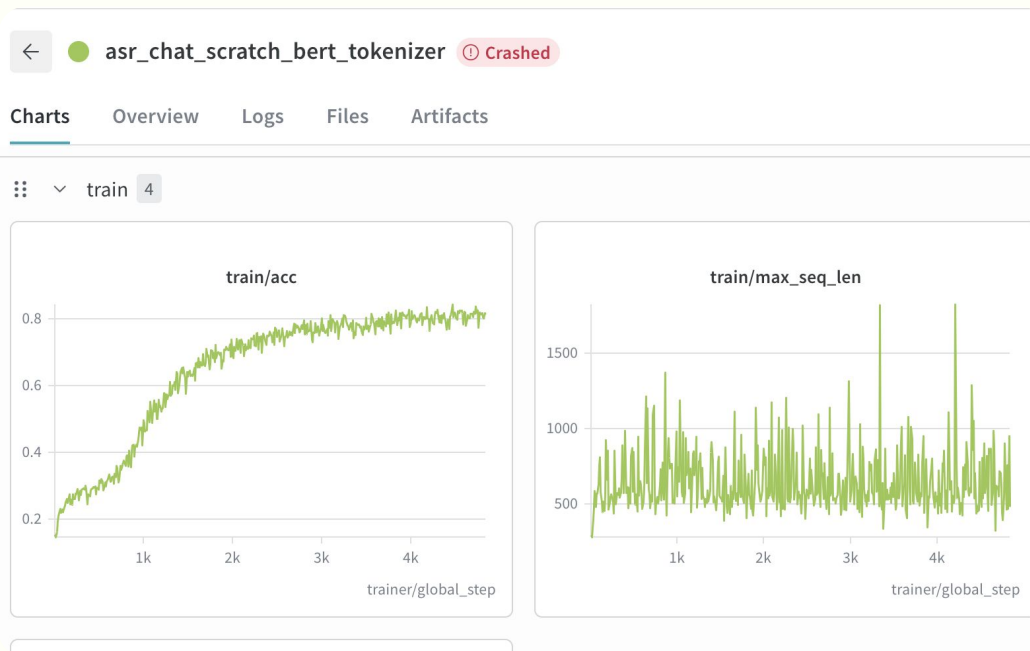
- **Unified dashboard:** Central repository for all your model metrics and predictions.
- **Lightweight:** No code changes required to integrate with Hugging Face.
- **Accessible:** Free for individuals and academic teams.
- **Secure:** All projects are private by default.
- **Trusted:** Used by machine learning teams at OpenAI, Toyota, Lyft, and more.

W&B

W&B works like GitHub for machine learning models. Save machine learning experiments to your private, hosted dashboard. Experiment with the confidence that W&B saves all versions of your models, no matter where you run your scripts.

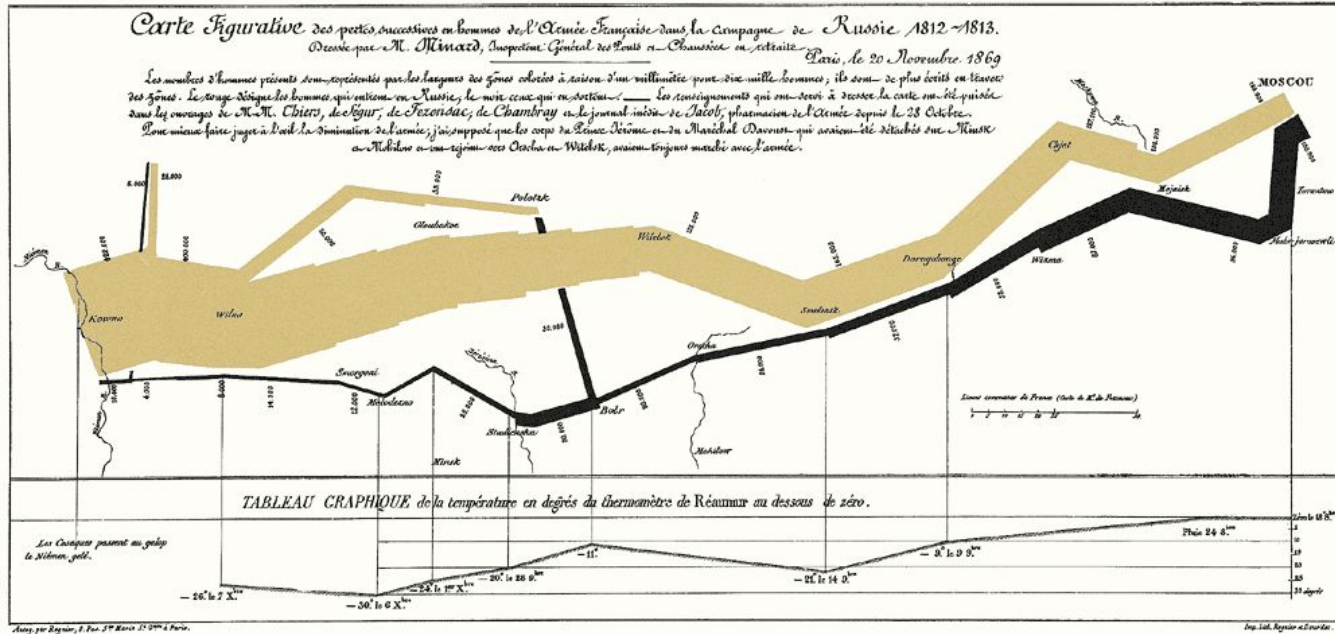
W&B lightweight integrations work with any Python script. Sign up for a free W&B account to start tracking and visualizing your models.

In the Hugging Face Transformers repository, W&B has instrumented the Trainer to automatically log training and evaluation metrics to W&B at each logging step.



Back to the basics

The art of visual storytelling



```
1 import seaborn as sns
2 from scipy.stats import linregress
3
4 df = sns.load_dataset("penguins").dropna(subset=["bill_depth_mm",
5 "body_mass_g"])
6 fit = linregress(df["bill_depth_mm"], df["body_mass_g"])
7 print(f"n={len(df)}   slope={fit.slope:.1f} g/mm   p=
8 {fit.pvalue:.1e}")
```

→ **n=342**

slope=-191.6 g/mm

p=2.3e-20

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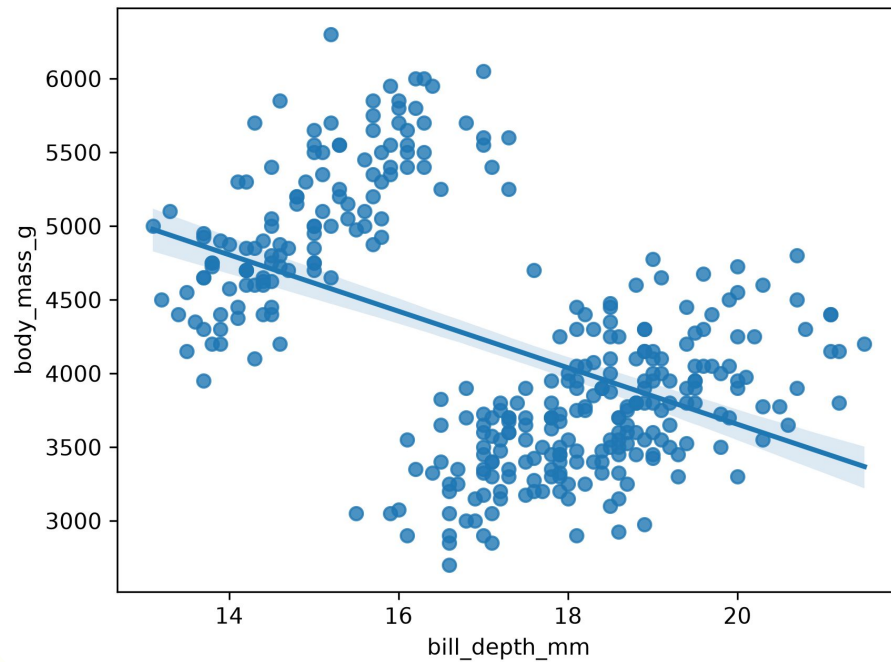
→ n=342

slope=-191.6 g/mm

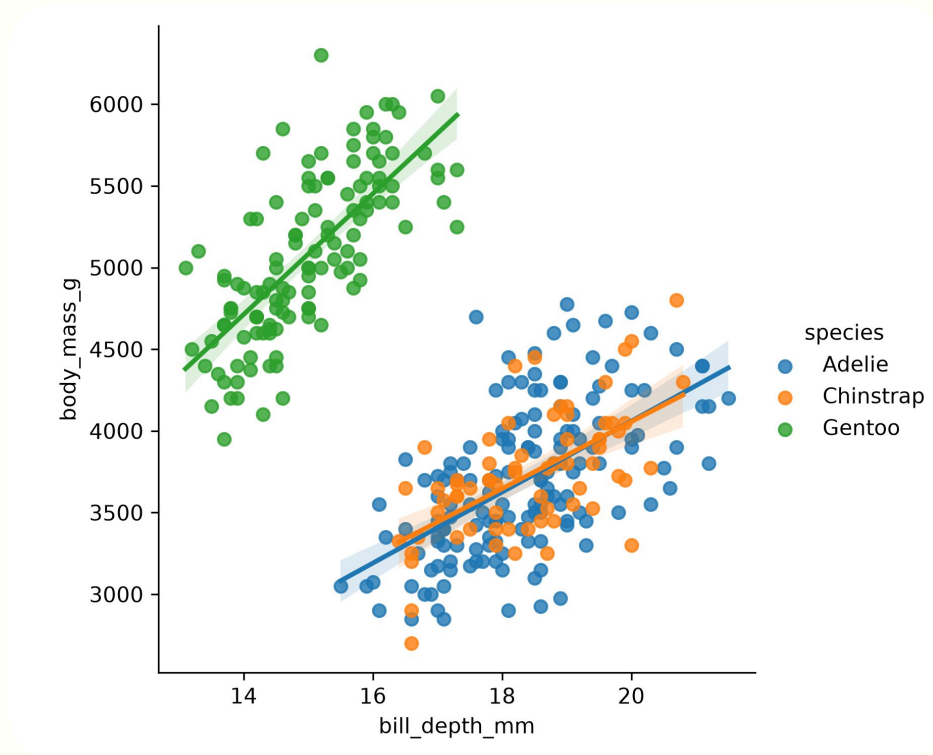
p=2.3e-20

Who here is good with p values?

Scatterplot is your best friend



Scatterplot is your best friend



Gentoo is also a penguin (gentoo.org)

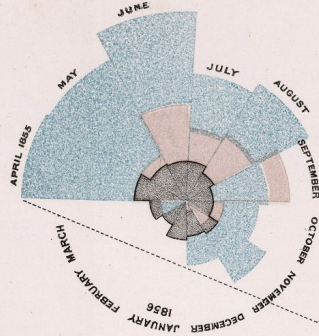


And a quick detour to a wonderful website

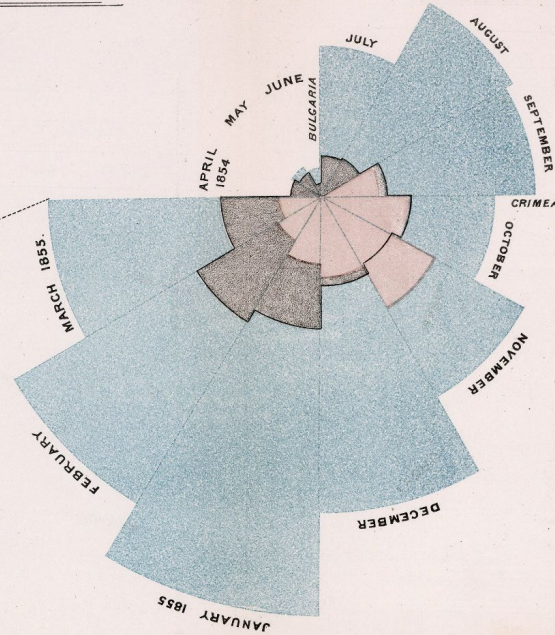
[Tylervigen - Spurious correlations](#)

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 TO MARCH 1856.

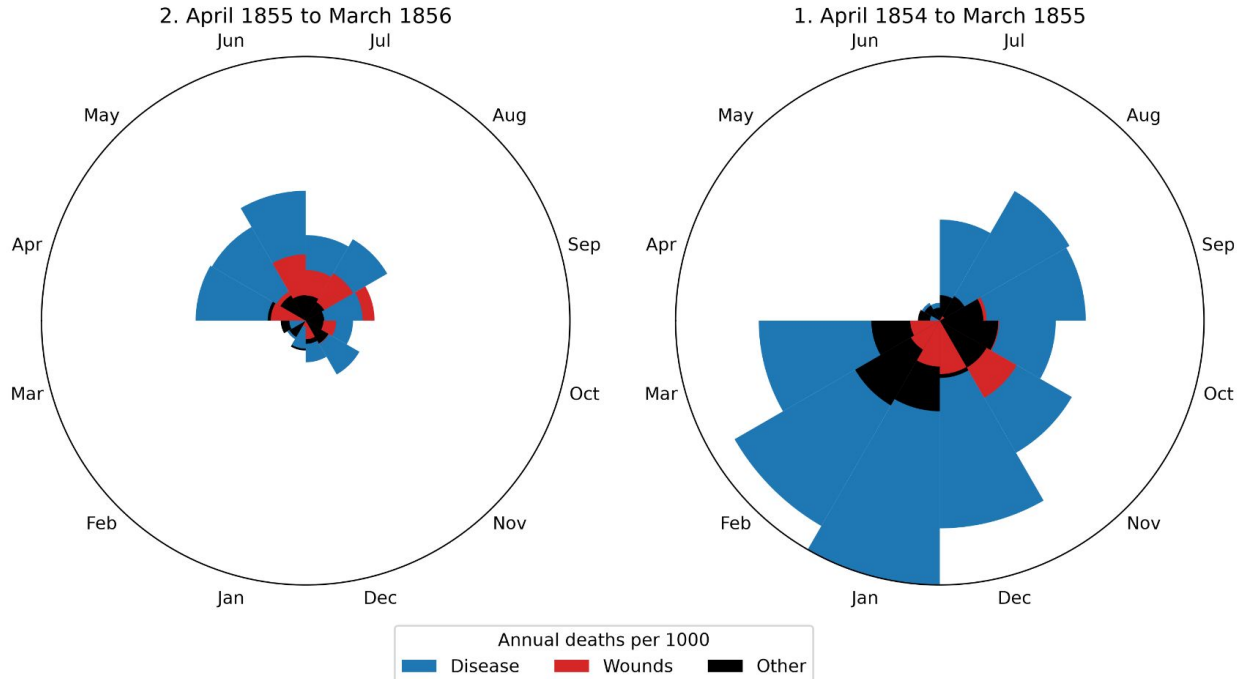


1.
APRIL 1854 TO MARCH 1855.

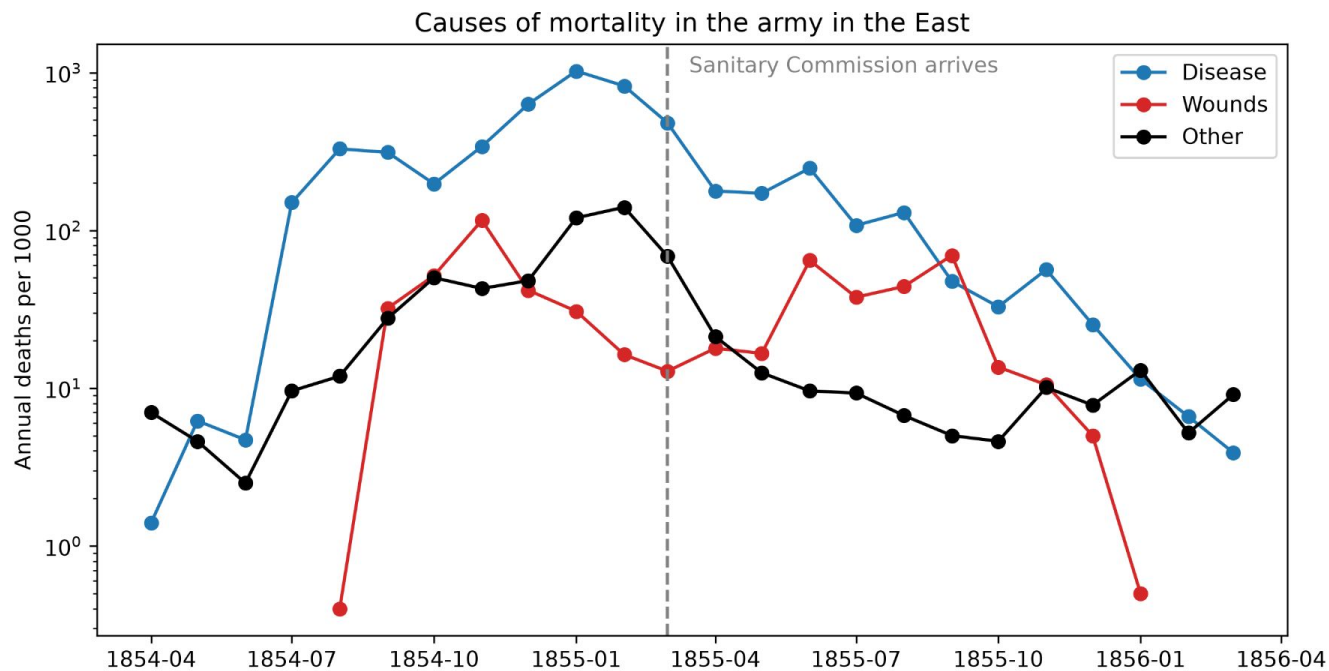


The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes.
The black line across the red triangle in Nov: 1854 marks the boundary of the deaths from all other causes during the month.
In October 1854, & April 1855, the black area coincides with the red;
in January & February 1855, the blue coincides with the black.
The entire areas may be compared by following the blue, the red & the black lines enclosing them.

Diagram of the causes of mortality in the army in the East



Alternative?



Takeaway

Think carefully about the questions you want to ask.

Think carefully about how you can present insightful answers.

Often, simple tooling is enough.

Important to mention here:

- Numpy, pandas
- Matplotlib, seaborn
- Scikit-learn, scipy.stats

Takeaway

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Nature

<https://www.nature.com> › review articles



Array programming with NumPy

by CR Harris · 2020 · Cited by 36644 — NumPy is a community-developed, open-source library, which provides a multidimensional Python array object along with array-aware functions that ...

Thank You

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